

PULSEGUARD TECHNOLOGIES INC. - STANDARD OPERATING PROCEDURE

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Title: TEST AND STAGING ENVIRONMENT MAINTENANCE PROTOCOLS
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Classification: Internal Restricted

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1. PURPOSE & OBJECTIVE

This Standard Operating Procedure (SOP) outlines the mandatory workflows, rules, and procedures for maintaining staging and test environments at PulseGuard. The goal is to maintain absolute environment health, verify data safety, ensure compliance with customer privacy regulations, and prevent staging configuration drifts that might invalidate pre-release verification metrics.

2. SCOPE

This procedure governs the configuration, data setup, cleanup routines, access, and freeze periods of all staging, development, sandbox, and automated testing environments owned or operated by PulseGuard Technologies Inc. It applies to all QA, Engineering, DevOps, and Product management staff.

3. ROLES AND RESPONSIBILITIES

3.1. Database Administrator (DBA)

- Manages database seeding scripts and anonymization engines.
- Wipes and restores test databases according to standard schedules.

3.2. DevOps Engineer

- Provisions staging VM/container clusters.
- Resolves environment connectivity issues and resource overflows.

3.3. QA Lead

- Declares freeze periods and coordinates testing windows.
- Approves manual test database updates.

3.4. Security Auditor

- Audits staging environments to confirm no raw PII exists.

4. DEFINITIONS AND REFERENCES

- Database Seeding: The process of populating databases with a standard set of test data before test runs.
- Anonymization: The process of scrubbing or replacing personal data (PII) with randomized parameters to protect customer privacy.
- Environment Freeze: A period where staging config updates, package updates, and hotfixes are suspended.
- SOP-QA-001: Mandatory Software Release Testing Protocols (reference).

5. STAGING AND TESTING DATA PROTECTION MANDATES

Staging and testing databases must strictly follow these rules:

5.1. Database Seeding Protocol

- Test environments must be seeded using randomized, mock, or fully anonymized datasets.
- Under no circumstances should raw, production customer data or files be imported into test/staging databases. This is a severe compliance and privacy violation.

- 5.2. Anonymization Requirements
- In the event that production datasets are needed for troubleshooting, the data must pass through the automated anonymization pipeline before import. All emails, phones, names, and keys must be scrambled.

6. ENVIRONMENT MAINTENANCE AND CLEANUP SCHEDULING

Testing environments must be regularly cleaned to avoid storage overhead:

- 6.1. Weekly Reset Schedule
- Every Friday at 6:00 PM UTC, an automated cleanup script must be executed by the orchestrator.
 - This script is configured to reset staging databases to seed templates and completely wipe all temporary storage directories and test files.
- 6.2. Drifting Verification
- Staging config files must be audited weekly against production.

7. COLD DEPLOYMENT AND ENVIRONMENT FREEZE PERIODS

To ensure stability before releasing major features, freezes are mandatory:

- 7.1. Freeze Timelines
- An environment freeze is declared 48 hours prior to any major public release.
 - During the freeze period, no database schema edits, API configurations, third-party updates, or hotfixes may be deployed to staging.
- 7.2. Override Procedures
- Freeze bypass requests require written sign-off from the QA Director.

8. PROCEDURAL STEP-BY-STEP FLOW

- Step 1: Staging environment is created or verified.
- Step 2: Friday 6 PM cleanup runs, resetting the environment to defaults.
- Step 3: DBA executes the mock seeding script, populating dummy user profiles.
- Step 4: Testers utilize the clean seeded environment for manual and auto tests.
- Step 5: 48 hours before release, QA Lead declares the Staging Freeze.
- Step 6: Release candidate is validated under static staging configurations.
- Step 7: Deployment to production occurs.
- Step 8: Freeze is lifted, resuming normal staging modifications.

9. TROUBLESHOOTING & ESCALATION PROCEDURES

- 9.1. Production Data Leaked in Staging
- If a user spots raw customer data in staging, they must isolate the DBA server immediately and flag the incident as Grade-1.
- 9.2. Escalation Contact Path
- First Level: Department QA Lead (Diana Patel).
 - Second Level: Sneha Deshmukh (VP of Engineering / Escalation Manager).
 - DevOps Escalation: devops-alerts@policypulse.dev.

10. REVISION HISTORY LOG

Version	Date	Author	Description of Change

v1.0	2024-06-11	D. Patel	Initial staging environment SOP.
v2.0	2025-01-05	D. Patel	Mandated weekly Friday cleanup
scripts.			
v2.2	2026-01-15	D. Patel	Added strict anonymization rules.
